

RESEARCH PROJECT GUIDELINES

ECON301—ECONOMETRICS I

PROF. ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

The project is due on Sunday, May 8, 2022, before 23:59 through Moodle. You can do your project with a team of *at most* five students. All required files (see below) should be submitted through the Moodle link. Follow the guidelines below when writing your project. The R functions `Fit1`, `WaldLinear`, and `WaldG` are provided by me during the lectures, and you can find them in Moodle.

Guidelines

1. Carefully formulate a question of interest (QE).
2. Use some kind of theory to give theoretical context to the QE.
3. Define some relevant variables that have to do with the QE. Separate them in dependent and explanatory variables.
4. Propose a Best Linear Predictor (BLP) and state the theoretical signs of the coefficients.
5. Find a database relevant for estimating the BLP of step 4. Use cross section data and put it as a *.csv file.
6. Consider if it is of interest to model the log of the dependent variable.
7. Do some exploratory plots of your data, particularly scatter plots.
8. Fit the BLP using the `Fit1` function.
9. Check the correspondence of the theoretical signs with those of the estimation procedure.
10. Make the boxplot of residuals and check if there are some outlying observations.
11. Iterate to exclude the outliers from the estimation procedure.
12. Compute the test of significance of the model. Use the function `WaldG`.
13. Check if there is evidence of multicollinearity and quasi multicollinearity.
14. Interpret the results of the estimation procedure in terms of the QE. You will need to do statistical inference regarding the parameters you are estimating, that is, you will need to carry out relevant hypothesis tests, or construct confidence intervals and interpret them. Depending on what makes sense for your particular QE, you may need to carry out inference on single coefficients, a linear combination of coefficients, or general linear hypothesis about several coefficients. You will need to determine what makes sense and it is interesting depending on your QE. Use the functions `Fit1`, `WaldLinear`, and `WaldG` for this.
15. Estimate at least one interaction effect and make the interaction plot. Preferably that it is significant.
16. Estimate at least one quadratic effect and make the quadratic effect plot. Preferably that it is significant.
17. Choose a meaningful set of explanatory variables and make the corresponding exclusion test. Use the function `WaldG`.
18. Write a report of professional quality without typos, and excellent presentation. Tables and figures should be numbered. With coherence between text, figures and tables. Write your conclusion pertaining the QE backed by the statistical analysis of your data (confidence intervals and test of hypothesis). Your document should be structured using the following sections:
 - i. Introduction and Motives for the Research.
 - ii. Literature review.
 - iii. Data, Model and Pretests.
 - iv. Results and Inference.
 - v. Conclusions.

vi. Bibliographic References

All bibliographic references used in your project should be properly cited, both in text and in the Bibliographic References section.

19. Use a maximum of ten pages with 12 pt Times New Roman (or similar) font.
20. Turn in the following files: your report in *.pdf file, your raw data in *.csv file, your R program in *.R file.

Grading Rubric

The research project grade is determined using the rubric in table 1. As indicated in the syllabus, the research project is worth 20% of your final grade. vi

Components	Weight
Introduction and motives for the research	15%
Integration of the literature review into the paper	15%
Model and pretests	20%
Results and inference	20%
Conclusions	15%
Grammar, writing quality	15%

Table 1: Grading rubric for ECON301 Research Project